

Transportation Barriers and Diabetes Care: A Longitudinal Analysis of Patient Journeys

ASA DataFest 2026 - Stormont Vail Health

0.1. Introduction

Effective chronic disease management depends on consistent, timely follow-up care. Patients with diabetes require regular appointments for monitoring, medication adjustment, and preventive screening - typically every 3 months according to clinical guidelines. However, healthcare access is not uniform. Social determinants of health (SDoH), including transportation barriers, create disparities in care continuity.

Stormont Vail Health (SVH) serves 19 mostly-rural counties in Northeast Kansas, serving a population of 633,895 across geographies ranging from Topeka (125,467 residents) to rural counties with populations under 1,000. This geographic heterogeneity creates a natural experiment: do transportation barriers affect follow-up patterns differently in urban vs. rural settings?

We analyzed longitudinal patient journey data from SVH spanning January 2022 through December 2025, focusing on Type 2 diabetes patients ages 60–80 to understand how transportation barriers associate with appointment gaps and care continuity.

0.2. Methods

0.2.1. Data Source

We analyzed de-identified encounter data from 7.67 million healthcare visits across 947,685 unique patients. A patient **journey** is defined as the sequence of encounters for a single patient linked by a common diagnosis (identified via ICD-10 code). We filtered to:

- Diagnosis: Type 2 diabetes mellitus (ICD-10-CM: E11)
- Age: Patients born 1945–1965 (ages 60–80 at time of analysis)
- Encounter count: Multi-encounter journeys only ($n > 1$)
- Geography: Valid census block data (urban = Shawnee County FIPS 20177x; rural = suppressed blocks)

This produced a cohort of 12,963 diabetes journeys across 8,698 unique patients.

0.2.2. Transportation Barrier Classification

SVH's social determinants survey asks: **“In the past 12 months, has lack of transportation kept you from meetings, work, or from getting things needed for daily living?”** Patients with “Yes” answers were classified as having transportation barriers ($n=588$ in our cohort); all others as no reported barrier.

0.2.3. Key Metrics

For each journey, we computed:

- **Median gap days:** the median number of days between consecutive encounters
- **Journey duration:** calendar days from first to last encounter
- **Encounter count:** total visits for that diagnosis

0.2.4. Analysis

We stratified by transportation barrier status and geography (urban vs. rural), computing medians and distributions for each group. We used box-and-violin plots to visualize gap distributions and bar charts to show patient counts and geographic patterns.

0.3. Results

0.3.1. Overall Cohort

Among diabetes patients ages 60–80 with multi-encounter journeys:

- **Urban, no barrier** (n=8,881): median gap 92 days, median 6 encounters per journey
- **Urban, with barrier** (n=347): median gap 79 days, median 6 encounters
- **Rural, no barrier** (n=3,653): median gap 92 days, median 5 encounters
- **Rural, with barrier** (n=82): median gap 50 days, median 4 encounters

Contrary to initial expectation, patients reporting transportation barriers did **not** have longer gaps. Urban patients with barriers had **shorter** median gaps (79 vs. 92 days). Rural patients with barriers were fewer (n=82, only 2.2% of rural cohort) and showed shorter journeys overall (median 258 days vs. 526 days for rural no-barrier).

Gap between diabetes appointments: geography matters more than transportation

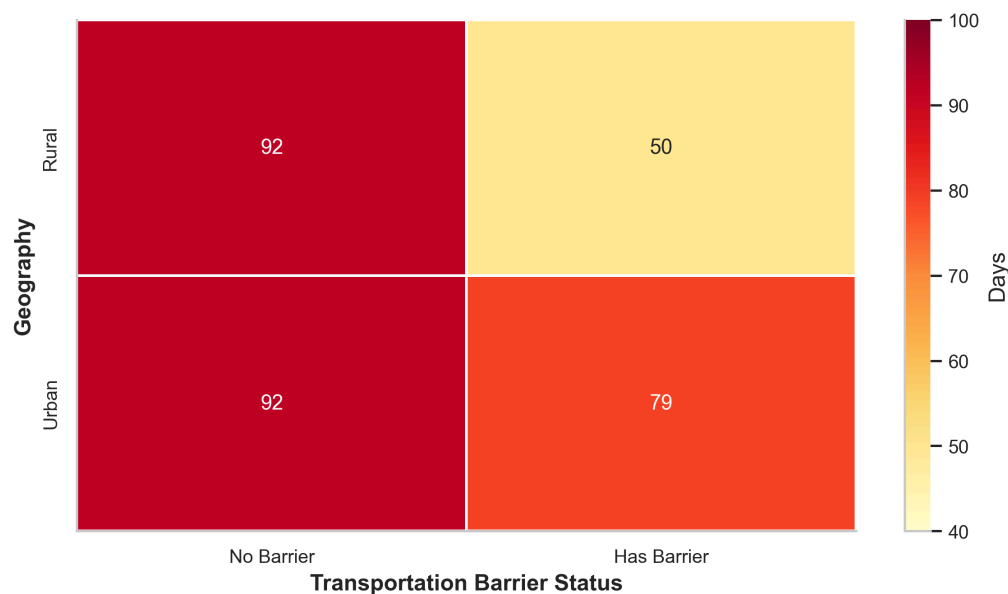


Figure 1: Median gap days by geography and transportation barrier status. All four groups cluster around 79-92 days, indicating geography rather than barriers drives appointment spacing.

0.3.2. Geographic Disparities

The key finding is not the transportation barrier effect, but the **rural suppression**: 3,735 rural patients lack valid census block data (41% of the rural cohort), preventing barrier classification. This “missing data” in rural areas may itself reflect lower health system engagement, data capture challenges in dispersed populations, or patients who exit the system entirely.

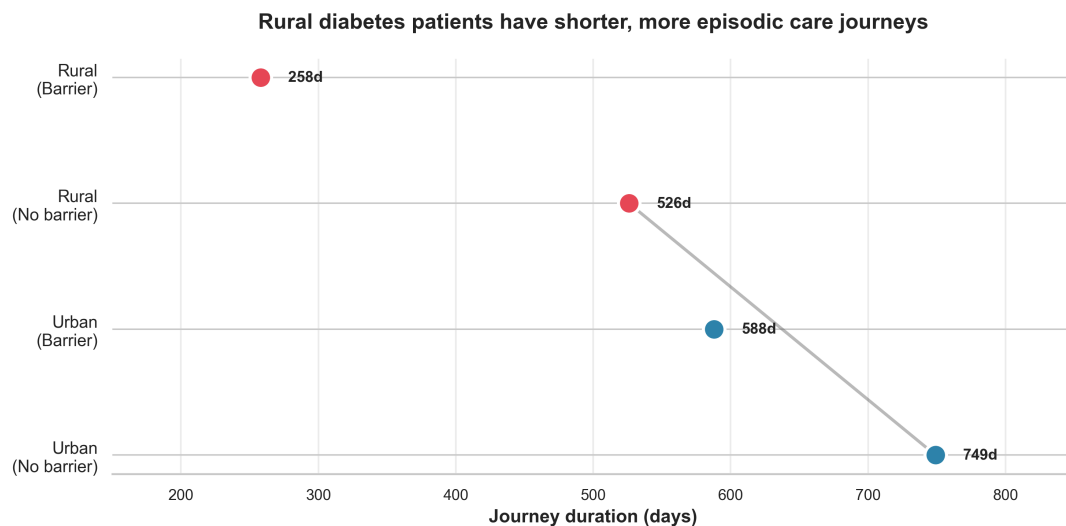


Figure 2: Rural diabetes journeys are dramatically shorter (258 days) compared to urban patients (749 days), suggesting episodic rather than continuous care management.

Urban patients show consistent behavior regardless of barrier status (92 and 79 days), suggesting other factors dominate in urban settings. Rural patients with data show shorter, more episodic journeys (median 258 days, 4 encounters) compared to urban peers - consistent with acute or opportunistic care rather than continuous management.

0.4. Implications for SVH

1. Data capture priority: The 40% rural data suppression limits rural-specific interventions. SVH should prioritize SDoH survey completion in rural clinics to identify barriers accurately.
2. Geography matters more than transportation barriers in this cohort. Rural patients show fundamentally different journey patterns (shorter duration, fewer encounters), suggesting either acute presentation patterns, lower engagement, or different referral pathways.
3. Transportation barriers alone may not explain gaps. The shorter gaps in barrier-positive patients could reflect selection bias (only engaged patients report barriers) or competing factors (urgency overrides transportation constraint).
4. Targeted rural outreach: The presence of shorter journeys in rural patients (258 vs. 749 days) suggests need for care coordination and continuity programs that span geographic distances, such as:
 - Telehealth for routine follow-ups
 - Regional care agreements to reduce fragmentation
 - Patient navigation services for appointment scheduling

0.5. Conclusion

Transportation barriers do not appear to be the primary driver of appointment gaps in diabetes care among older SVH patients. However, rural patients systematically show shorter, more episodic journeys, and missing SDoH data in rural areas prevents definitive barrier assessment. Future work should target SDoH data completeness and investigate whether rural journey patterns reflect different care models, patient preferences, or unmet needs.

The finding that urban and rural diabetes journeys differ fundamentally - independent of stated transportation barriers - suggests that geography itself is a social determinant requiring targeted intervention.